

HandsOn-4

- Implement componentDidMount() hook
- Implementing componentDidCatch() life cycle hook.

Implementation

→ App.js

// src/App.js

```
import React from 'react';  
import Posts from './Posts';
```

```
function App() {  
  return (  
    <div className="App">  
      <Posts />  
    </div>  
  );  
}  
export default App;
```

→ Errorcomponent.js

// src/ErrorComponent.js

```
import React from 'react';
```

```
function ErrorComponent() {  
  throw new Error("Deliberate error to test componentDidCatch");  
  
export default ErrorComponent;
```

→ Post.js

```
// src/Post.js

class Post {

  constructor(userId, id, title, body) {

    this.userId = userId;

    this.id = id;

    this.title = title;

    this.body = body;

  }

}

export default Post;
```

→ Posts.js

```
// src/Posts.js

import React, { Component } from 'react';
import Post from './Post';
import ErrorComponent from './ErrorComponent';

class Posts extends Component {

  constructor(props) {

    super(props);

    this.state = {

      posts: [],

      hasError: false

    };

  }

  // Fetch posts from API

  loadPosts = () => {
```

```

fetch('https://jsonplaceholder.typicode.com/posts')
  .then((response) => response.json())
  .then((data) => {
    const postList = data.map(
      (post) => new Post(post.userId, post.id, post.title, post.body)
    );
    this.setState({ posts: postList });
  })
  .catch((error) => {
    console.error("Error in fetch:", error);
    this.setState({ hasError: true });
  });
};

```

// Call fetch after component mounts

```

componentDidMount() {
  this.loadPosts();
}

```

// Catch rendering errors from children

```

componentDidCatch(error, info) {
  console.error("componentDidCatch caught an error:", error, info);
  this.setState({ hasError: true });
}

```

```

render() {
  if (this.state.hasError) {
    return (
      <div style={{ color: 'red', padding: '20px' }}>
        <h3> ⚠️ Oops! Something went wrong.</h3>

```

```

    <p>The posts could not be loaded due to an internal error.</p>
  </div>

  );
}

return (
  <div style={{ padding: '20px' }}>
    <h2>Blog Posts</h2>

    {/* ErrorComponent removed */}

    {this.state.posts.map((post) => (
      <div key={post.id} style={{ marginBottom: '20px' }}>
        <h3>{post.title}</h3>
        <p>{post.body}</p>
        <hr />
      </div>
    ))}
  </div>

  );
}

}

export default Posts;

```

(correct render)

```

render() {
  if (this.state.hasError) {
    return (
      <div style={{ color: 'red', padding: '20px' }}>
        <h3> ⚠️ Oops! Something went wrong.</h3>

```

```
    <p>The posts could not be loaded due to an internal error.</p>
  </div>

);
}

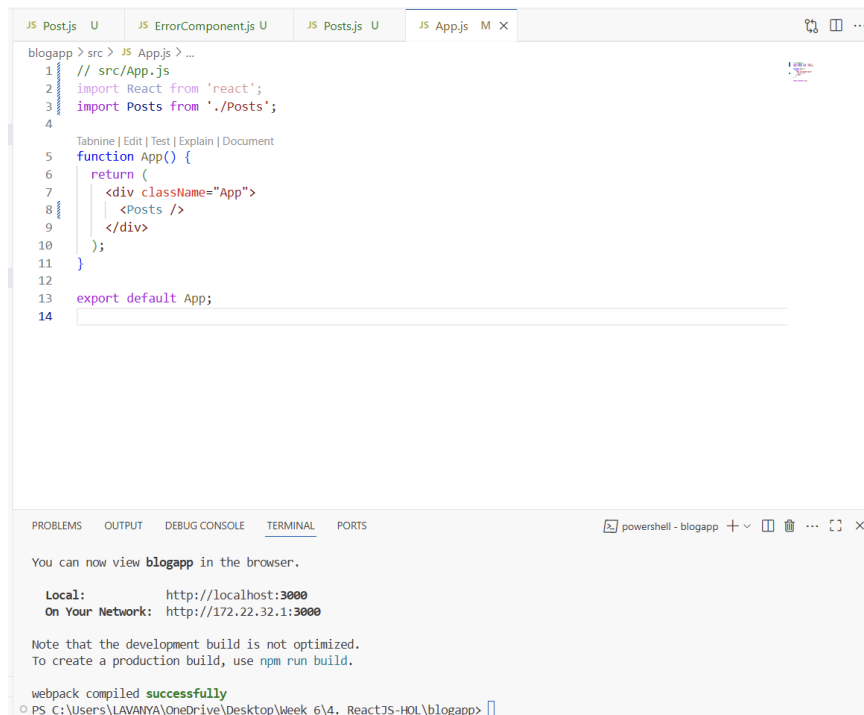
return (
  <div style={{ padding: '20px' }}>
    <h2>Blog Posts</h2>

    {/* ErrorComponent removed */}

    {this.state.posts.map((post) => (
      <div key={post.id} style={{ marginBottom: '20px' }}>
        <h3>{post.title}</h3>
        <p>{post.body}</p>
        <hr />
      </div>
    ))}
  </div>

);
}
```

Output



The screenshot shows the VS Code editor with the `App.js` file open. The code defines a functional component `App` that renders a `<div>` containing a `<Posts>` component. The terminal at the bottom shows the command `blogapp` being run, which outputs the local and network URLs, a note about the development build, and a successful webpack compilation message.

```
blogapp > src > JS App.js > ...
1 // src/App.js
2 import React from 'react';
3 import Posts from './Posts';
4
5 function App() {
6   return (
7     <div className="App">
8       <Posts />
9     </div>
10   );
11 }
12
13 export default App;
14
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

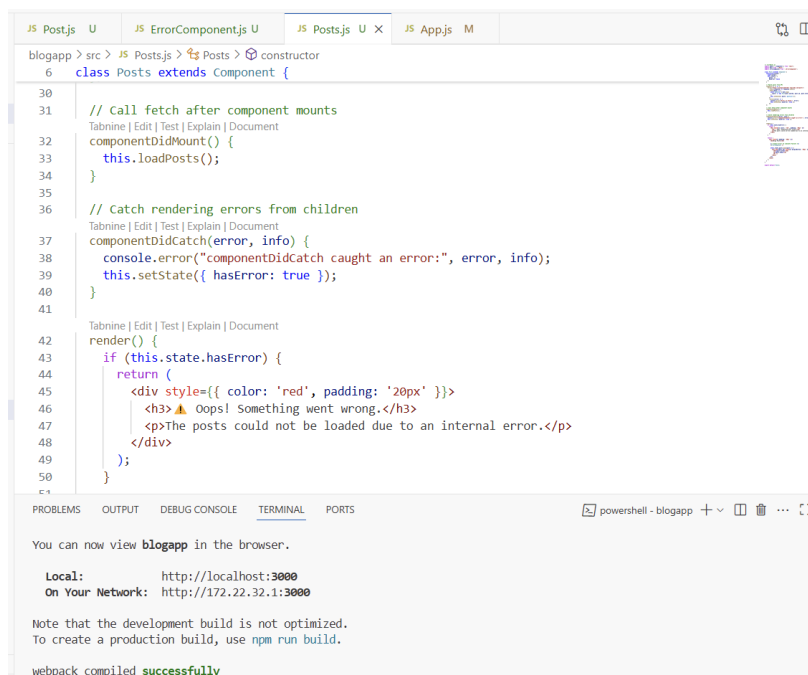
You can now view **blogapp** in the browser.

Local: http://localhost:3000
On Your Network: http://172.22.32.1:3000

Note that the development build is not optimized.
To create a production build, use `npm run build`.

webpack compiled **successfully**

PS C:\Users\LAANIYA\OneDrive\Desktop\Week 6\4. ReactJS-HOL\blogapp>



The screenshot shows the VS Code editor with the `Posts.js` file open. The code defines a class component `Posts` that extends `Component`. It includes `componentDidMount` and `componentDidCatch` methods, and a `render` method that displays an error message if an error occurs. The terminal at the bottom shows the command `blogapp` being run, which outputs the local and network URLs, a note about the development build, and a successful webpack compilation message.

```
blogapp > src > JS Posts.js > Posts > constructor
6 class Posts extends Component {
30
31 // Call fetch after component mounts
32 componentDidMount() {
33   this.loadPosts();
34 }
35
36 // Catch rendering errors from children
37 componentDidCatch(error, info) {
38   console.error("componentDidCatch caught an error:", error, info);
39   this.setState({ hasError: true });
40 }
41
42 render() {
43   if (this.state.hasError) {
44     return (
45       <div style={{ color: 'red', padding: '20px' }}>
46         <h3>⚠️ Oops! Something went wrong.</h3>
47         <p>The posts could not be loaded due to an internal error.</p>
48       </div>
49     );
50   }
51 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

You can now view **blogapp** in the browser.

Local: http://localhost:3000
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Blog Posts

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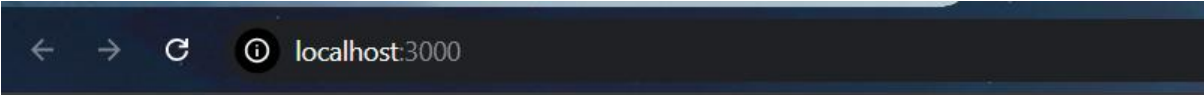
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 **Oops! Something went wrong.**

The posts could not be loaded due to an internal error.

```

render() {
  if (this.state.hasError) {
    return (
      <div style={{ color: 'red', padding: '20px' }}>
        <h3>⚠️ Oops! Something went wrong.</h3>
        <p>The posts could not be loaded due to an internal error.</p>
      </div>
    );
  }

  return (
    <div style={{ padding: '20px' }}>
      <h2>Blog Posts</h2>

      {/* ErrorComponent removed */}

      {this.state.posts.map((post) => (
        <div key={post.id} style={{ marginBottom: '20px' }}>
          <h3>{post.title}</h3>
          <p>{post.body}</p>
          <hr />
        </div>
      ))}
    </div>
  );
}

```

